

Computer Networks — Lecture 02

Circuit Switching, Network of Networks, Delay and Loss (1.3.2 - 1.4)

المحاضرة 2 — تبديل الدوائر وشبكة الشبكات والتأخير والفقد

Part 1: Multiple Choice Questions

1. In a circuit-switched network, what happens to the resources needed along a path (buffers, link transmission rate)?

في شبكات تبديل الدوائر، ماذا يحدث للموارد اللازمة على المسار (ال buffers ومعدل الإرسال)؟

- A) They are shared on demand with no reservation
- B) They are reserved only for the first packet of the session
- C) They are reserved for the duration of the communication session
- D) They are never allocated to a specific session

✓ **الإجابة الصحيحة:** C) They are reserved for the duration of the communication session

► **ليه صح:** في تبديل الدوائر تُحجز الموارد طوال مدة جلسة الاتصال. الاختيار الأول يصف تبديل الحزم (packet switching) حيث لا يوجد حجز.

● Ref: Lecture 02 — Section 1.3.2

2. In the restaurant analogy, which type of restaurant corresponds to circuit switching?

في تشبيه المطاعم، أي مطعم يقابل تبديل الدوائر؟

- A) The restaurant that requires reservations
- B) A restaurant that reserves tables only for tier-1 customers
- C) The restaurant that neither requires nor accepts reservations
- D) A restaurant that serves food only to groups

✓ **الإجابة الصحيحة:** A) The restaurant that requires reservations

► **ليه صح:** المطعم الذي يشترط الحجز يقابل تبديل الدوائر: تحجز مسبقًا ثم تجلس فورًا عند الوصول. المطعم بلا حجز يقابل packet switching حيث قد تنتظر.

● Ref: Lecture 02 — Section 1.3.2

3. How is a circuit in a link implemented in circuit-switched networks?

كيف تُنفَّذ الدائرة (circuit) داخل الرابط في شبكات تبديل الدوائر؟

- A) With forwarding tables that map addresses to links
- B) With either frequency-division multiplexing (FDM) or time-division multiplexing (TDM)
- C) With peering links between ISPs
- D) With store-and-forward buffering at each router

✓ **الإجابة الصحيحة:** B) With either frequency-division multiplexing (FDM) or time-division multiplexing (TDM)

► **ليه صح:** الدائرة تُنفَّذ إما بال FDM أو بال TDM. ال store-and-forward وجداول التوجيه من خصائص تبديل الحزم.

● Ref: Lecture 02 — Section 1.3.2

4. Why have critics argued that packet switching is not suitable for real-time services such as telephone calls?

لماذا انتقد البعض استخدام packet switching في الخدمات الفورية مثل المكالمات؟

- A) Because of its variable and unpredictable end-to-end delays, mainly due to queuing delays
- B) Because it requires a reservation before each call
- C) Because it cannot divide messages into smaller chunks
- D) Because it needs a dedicated link per user

✓ **الإجابة الصحيحة:** A) Because of its variable and unpredictable end-to-end delays, mainly due to queuing delays

► **ليه صح:** لأن التأخير من طرف إلى طرف متغير وغير متوقع بسبب تأخيرات الطابور (queuing). الحجز وتخصيص رابط لكل مستخدم من صفات تبديل الدوائر.

● Ref: Lecture 02 — Section 1.3.2

5. Which are the two main arguments of packet switching proponents over circuit switching?

ما حجتا مؤيدي packet switching الأساسيتان على تبديل الدوائر؟

- A) It needs no buffers, and it never loses packets
- B) It offers better sharing of transmission capacity, and it is simpler, more efficient, and less costly to implement
- C) It guarantees constant delay, and it reserves resources for each session
- D) It works only for real-time services, and it requires less bandwidth for each call

✓ **الإجابة الصحيحة:** B) It offers better sharing of transmission capacity, and it is simpler, more efficient, and less costly to implement

► **ليه صح:** المؤيدون يقولون إنه يتيح مشاركة أفضل للسعة وأنه أبسط وأكفأ وأقل تكلفة في التنفيذ. ضمان تأخير ثابت وحجز الموارد من مزايا تبديل الدوائر.

● Ref: Lecture 02 — Section 1.3.2

6. What is the trend in today's telecommunication networks, and how do telephone networks use packet switching?

ما اتجاه شبكات الاتصالات اليوم، وكيف تستخدم شبكات الهاتف تبديل الحزم؟

- A) Both are declining; telephone networks use packet switching only inside homes
 - B) The trend is toward circuit switching; telephone networks avoid packet switching completely
 - C) The trend is toward packet switching; telephone networks often use it for the expensive overseas portion of a call
 - D) The trend is toward packet switching; telephone networks use it only for the local portion of a call
- ✓ **الإجابة الصحيحة:** C) The trend is toward packet switching; telephone networks often use it for the expensive overseas portion of a call
- **ليه صح:** الاتجاه نحو packet switching، وشبكات الهاتف تستخدمه غالبًا في الجزء الدولي (overseas) الغالي من المكالمات. الاختيار الأخير يعكس الجزء المستخدم فيه.

● Ref: Lecture 02 — Section 1.3.2

7. Why is directly connecting every access ISP to every other access ISP not a viable design?

لماذا لا يصلح ربط كل access ISP مباشرة بكل access ISP آخر؟

- A) It would make packets travel through too many tier-1 ISPs
 - B) It would prevent customers from choosing a provider
 - C) It would need a single global transit ISP to charge all of them
 - D) It does not scale: each access ISP would need a separate link to each of hundreds of thousands of other access ISPs, which is far too costly
- ✓ **الإجابة الصحيحة:** D) It does not scale: each access ISP would need a separate link to each of hundreds of thousands of other access ISPs, which is far too costly
- **ليه صح:** لأن التصميم (mesh) لا يتوسع: كل ISP يحتاج رابطًا مستقلًا لمئات الآلاف من الـ ISPs الأخرى وهذا مكلف جدًا. وجود global transit ISP هو الحل البديل (Network Structure 1) وليس سبب الفشل.

● Ref: Lecture 02 — Section 1.3.3

8. In Network Structure 1 (a single global transit ISP), who pays whom?

في Network Structure 1 (global transit ISP واحد)، من يدفع لمن؟

- A) The global transit ISP pays each access ISP
 - B) Neither pays; connectivity is settlement-free
 - C) Each access ISP pays the global transit ISP, so the access ISP is the customer and the transit ISP is the provider
 - D) Access ISPs pay each other for the traffic they exchange
- ✓ **الإجابة الصحيحة:** C) Each access ISP pays the global transit ISP, so the access ISP is the customer and the transit ISP is the provider
- **ليه صح:** ال access ISP يدفع للـ global transit ISP فيكون زبونًا (customer) والآخر مزودًا (provider). الدفع المتبادل بدون مقابل (settlement-free) يخص ال peering وليس هذه العلاقة.

● Ref: Lecture 02 — Section 1.3.3

9. Why do access ISPs prefer Network Structure 2 (multiple competing global transit ISPs) over Network Structure 1?

لماذا تفضل الـ access ISPs بنية Network Structure 2 على Network Structure 1؟

- A) They no longer need to pay any provider
 - B) They can choose among competing providers according to pricing and services
 - C) They can connect directly to every other access ISP
 - D) Global transit ISPs no longer need to interconnect
- ✓ **الإجابة الصحيحة:** B) They can choose among competing providers according to pricing and services
- **ليه صح:** لأنها تستطيع الاختيار بين مزودين متنافسين حسب السعر والخدمة. ومع ذلك يجب أن تتصل الـ global transit ISPs ببعضها وإلا لن تتواصل الشبكات المتصلة بمزودين مختلفين.

● Ref: Lecture 02 — Section 1.3.3

10. In the multi-tier hierarchy (Network Structure 3), which ISP does a regional ISP pay?

في البنية الهرمية (Network Structure 3)، لمن يدفع الـ regional ISP؟

- A) The access ISPs it serves
 - B) The tier-1 ISP to which it connects
 - C) No one, because regional ISPs are at the top
 - D) Another regional ISP at the same level
- ✓ **الإجابة الصحيحة:** B) The tier-1 ISP to which it connects
- **ليه صح:** كل access ISP يدفع للـ regional ISP، وكل regional ISP يدفع للـ tier-1 ISP الذي يتصل به. الـ tier-1 هي التي لا تدفع لأحد لأنها في قمة الهرم.

● Ref: Lecture 02 — Section 1.3.3

11. Which statement about tier-1 ISPs is correct?

أي عبارة عن ال tier-1 ISPs صحيحة؟

- A) There are thousands of them, one in every city
- B) They pay regional ISPs for transit
- C) They never peer with each other
- D) They do not pay anyone, they peer with each other settlement-free, and none has a presence in every city of the world

✓ **الإجابة الصحيحة:** D) They do not pay anyone, they peer with each other settlement-free, and none has a presence in every city of the world

► **ليه صح:** ال tier-1 ISPs لا تدفع لأحد، وتتبادل الحركة فيما بينها بدون مقابل (peering)، وليس لأي منها وجود في كل مدينة، وعددها حوالي اثنتي عشرة. الاختيار الثاني خطأ في العدد.

● Ref: Lecture 02 — Section 1.3.3

12. What is a point of presence (PoP)?

ما هي نقطة التواجد (PoP)؟

- A) A group of one or more routers in the provider's network where customer ISPs can connect into the provider ISP
- B) A content provider's private data center
- C) A meeting point built by a third party where several ISPs peer
- D) A direct link between two ISPs at the same level

✓ **الإجابة الصحيحة:** A) A group of one or more routers in the provider's network where customer ISPs can connect into the provider ISP

► **ليه صح:** ال PoP مجموعة راوتر أو أكثر في شبكة المزود يتصل بها ال customer ISPs. الاختيار الأول يصف ال IXP، والثاني يصف ال peering link.

● Ref: Lecture 02 — Section 1.3.3

13. What is the benefit of multi-homing for an ISP?

ما فائدة ال multi-homing لأي ISP؟

- A) It stops paying its providers
- B) It becomes a tier-1 ISP
- C) It can bypass its customers' traffic
- D) It can continue to send and receive packets even if one of its providers fails

✓ **الإجابة الصحيحة:** D) It can continue to send and receive packets even if one of its providers fails

► **ليه صح:** ال multi-homing هو الاتصال بمزودين أو أكثر، وهذا يسمح بالاستمرار في الإرسال والاستقبال لو فشل أحد المزودين. لا يُعفي ISP من الدفع ولا يحوِّله إلى tier-1.

● Ref: Lecture 02 — Section 1.3.3

14. Why might two nearby ISPs at the same level of the hierarchy decide to peer?

لماذا قد يقرر ISP ان قريبان في نفس المستوى الاتصال ببعضهما (peering)؟

- A) To become each other's provider and charge each other
- B) To remove all of their customers from the network
- C) To obtain a presence in every city
- D) To connect their networks directly and avoid paying upstream intermediaries for the traffic exchanged between them

✓ **الإجابة الصحيحة:** D) To connect their networks directly and avoid paying upstream intermediaries for the traffic exchanged between them

► **ليه صح:** لتمر الحركة بينهما عبر الرابط المباشر بدل المرور عبر وسطاء في الأعلى، فتقل تكلفة الدفع. وال peering غالبًا settlement-free أي لا يدفع أحدهما للآخر.

● Ref: Lecture 02 — Section 1.3.3

15. What is an Internet exchange point (IXP)?

ما هو ال Internet exchange point (IXP)؟

- A) A private network that carries only a single company's traffic
- B) A group of routers inside a provider ISP where customers connect
- C) A meeting point, typically a stand-alone building with its own switches, where multiple ISPs can peer together
- D) A link that connects an access ISP to exactly one provider

✓ **الإجابة الصحيحة:** C) A meeting point, typically a stand-alone building with its own switches, where multiple ISPs can peer together

► **ليه صح:** ال IXP نقطة لقاء تُنشئها طرف ثالث، وغالبًا في مبنى مستقل بمحولاته، ليتصل فيها عدة ISPs ببعضهم. الاختيار الأول هو ال PoP.

● Ref: Lecture 02 — Section 1.3.3

16. What does Network Structure 5, which describes today's Internet, add on top of Network Structure 4?

ماذا تضيف Network Structure 5 (التي تصف إنترنت اليوم) فوق Network Structure 4؟

- A) Multi-homing
- B) A single global transit ISP
- C) Points of presence
- D) Content-provider networks

✓ **الإجابة الصحيحة:** D) Content-provider networks

► **ليه صح:** تضيف شبكات مزودي المحتوى (content-provider networks) مثل شبكة Google. ال PoPs وال multi-homing موجودة بالفعل في Network Structure 4.

● Ref: Lecture 02 — Section 1.3.3

17. Why does a content provider such as Google build its own private network?

لماذا يبني مزود محتوى مثل Google شبكته الخاصة؟

- A) It reduces its payments to upper-tier ISPs and gains greater control over how its services are delivered to end users
- B) It replaces the public Internet entirely for all users
- C) It eliminates the need to connect to any tier-1 ISP
- D) It lets it carry the traffic of other companies

✓ **الإجابة الصحيحة:** A) It reduces its payments to upper-tier ISPs and gains greater control over how its services are delivered to end users

► **ليه صح:** ليققل مدفوعاته للـ ISPs في المستويات العليا ويتحكم أكثر في طريقة وصول خدماته للمستخدمين. لكنه ما زال يتصل بـ tier-1 ISPs ويدفع لها عند الحاجة، وشبكته تحمل حركة خوادم Google فقط.

● Ref: Lecture 02 — Section 1.3.3

18. Which list contains the four main delays that make up total nodal delay?

أي قائمة تحتوي التأخيرات الأربعة الرئيسية المكونة للتأخير الكلي عند العقدة (nodal delay)؟

- A) Processing, queuing, transmission, and propagation
- B) Queuing, switching, propagation, and handshaking
- C) Processing, routing, transmission, and encoding
- D) Transmission, propagation, loss, and throughput

✓ **الإجابة الصحيحة:** A) Processing, queuing, transmission, and propagation

► **ليه صح:** التأخيرات الأربعة هي processing و queuing و transmission و propagation، ومجموعها هو nodal delay. الفقد (loss) وال throughput ليسا أنواع تأخير.

● Ref: Lecture 02 — Section 1.4.1

19. What is included in the processing delay, and what is its typical magnitude in high-speed routers?

ماذا يشمل تأخير المعالجة (processing delay) وما حجمه المعتاد في الراوترات السريعة؟

- A) Pushing all of the packet's bits onto the link; seconds
- B) Examining the packet header to decide where to direct it (and checking for bit-level errors); microseconds or less
- C) Waiting for earlier packets in the queue; milliseconds
- D) Propagating bits across the link; hundreds of milliseconds

✓ **الإجابة الصحيحة:** B) Examining the packet header to decide where to direct it (and checking for bit-level errors); microseconds or less

► **ليه صح:** يشمل فحص رأس الحزمة لتحديد وجهتها (وفحص أخطاء ال bits)، وحجمه ميكروثانية أو أقل. الانتظار في الطابور هو queuing، ودفع ال bits هو transmission.

● Ref: Lecture 02 — Section 1.4.1

20. A packet has length L bits and the link rate is R bits/sec. What is the transmission delay?

حزمة طولها L bit ومعدل الرابط R bit/sec. ما تأخير الإرسال (transmission delay)؟

- A) $L \times R$
- B) d/s
- C) R/L
- D) L/R

✓ الإجابة الصحيحة: D) L/R

► **ليه صح:** تأخير الإرسال L/R وهو الزمن اللازم لدفع كل bits الحزمة إلى الرابط. d/s هو قانون تأخير الانتشار (propagation delay).

● Ref: Lecture 02 — Section 1.4.1

21. The propagation delay is d/s . What do d and s represent, and what is the typical range of s?

تأخير الانتشار هو d/s . ما معنى d و s، وما المدى المعتاد لـ s؟

- A) d is the number of packets and s is the arrival rate, about 2 to 3 packets/sec
- B) d is the packet length and s is the link rate, about 10 to 100 Mbps
- C) d is the distance between the two routers and s is the propagation speed of the link, about 2×10^8 to 3×10^8 m/s
- D) d is the number of routers and s is the queue size, about 1 to 10 packets

✓ الإجابة الصحيحة: C) d is the distance between the two routers and s is the propagation speed of the link, about 2×10^8 to 3×10^8 m/s

► **ليه صح:** الـ d هي المسافة بين الراوترين و s سرعة الانتشار في الرابط وتتراوح بين 2×10^8 و 3×10^8 م/ث (قريبة من سرعة الضوء). الطول والمعدل يدخلان في تأخير الإرسال L/R .

● Ref: Lecture 02 — Section 1.4.1

22. Ten packets arrive at an empty queue at the same time. How much queuing delay do the first and the last transmitted packets suffer?

وصلت 10 حزم إلى طابور فارغ في نفس اللحظة. كم تأخير الطابور للحزمة الأولى والأخيرة؟

- A) The first suffers no queuing delay; the last waits for the other nine packets to be transmitted
- B) Both suffer zero queuing delay
- C) The first waits for nine packets; the last suffers none
- D) Both wait exactly the same amount of time

✓ الإجابة الصحيحة: A) The first suffers no queuing delay; the last waits for the other nine packets to be transmitted

► **ليه صح:** الأولى تُرسل مباشرة بلا انتظار، والأخيرة تنتظر الحزم التسع قبلها. هذا يبين أن تأخير الطابور يتغير من حزمة إلى أخرى بعكس باقي التأخيرات.

● Ref: Lecture 02 — Section 1.4.2

23. What happens when the traffic intensity $\lambda a/R$ is greater than 1?

ماذا يحدث عندما تكون شدة الحركة (traffic intensity) $\lambda a/R$ أكبر من 1؟

- A) The queue stays empty and the delay is zero
 - B) Packets arrive periodically with no delay
 - C) Bits arrive faster than they can be transmitted, so the queue grows without bound and the queuing delay approaches infinity
 - D) The queuing delay stays constant
- ✓ **الإجابة الصحيحة:** C) Bits arrive faster than they can be transmitted, so the queue grows without bound and the queuing delay approaches infinity
- **ليه صح:** عندها يفوق معدل وصول الـ bits معدل إرسالها فيكبر الطابور بلا حد ويقترب التأخير من اللانهاية، لذلك القاعدة الذهبية: صمم النظام بحيث لا تزيد الشدة عن 1.

● Ref: Lecture 02 — Section 1.4.2

24. Suppose there are $N-1$ routers between source and destination, the network is uncongested, and every node has the same d_{proc} , rate R , and d_{prop} . What is the end-to-end delay?

إذا وُجد $N-1$ راوتر بين المصدر والوجهة والشبكة غير مزدحمة وكل عقدة لها نفس d_{proc} و R و d_{prop} ، فما التأخير من طرف إلى طرف؟

- A) $N \times d_{\text{queue}} + L/R$
 - B) $N (d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$, where $d_{\text{trans}} = L/R$
 - C) $(N-1) (d_{\text{proc}} + d_{\text{prop}})$ only
 - D) $d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}}$ (a single node's delay)
- ✓ **الإجابة الصحيحة:** B) $N (d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$, where $d_{\text{trans}} = L/R$
- **ليه صح:** التأخيرات تتراكم عبر N رابط فيكون $d(\text{end-to-end}) = N(d_{\text{proc}} + d_{\text{trans}} + d_{\text{prop}})$ مع $d_{\text{trans}} = L/R$. وهي تعميم للمعادلة 1.1 التي أهملت تأخيري المعالجة والانتشار.

● Ref: Lecture 02 — Section 1.4.2

25. What is the relationship between traffic intensity and packet loss at a router with a finite buffer?

ما العلاقة بين شدة الحركة وفقد الحزم في راوتر ذي buffer محدود؟

- A) The fraction of lost packets increases as the traffic intensity increases
 - B) Packet loss occurs only when the traffic intensity is exactly zero
 - C) Packet loss is independent of the traffic intensity
 - D) Packet loss decreases as the traffic intensity increases
- ✓ **الإجابة الصحيحة:** A) The fraction of lost packets increases as the traffic intensity increases
- **ليه صح:** كلما زادت شدة الحركة زادت نسبة الحزم المفقودة. والحزمة المفقودة قد تُعاد إرسالها من طرف إلى طرف لضمان وصول البيانات.

● Ref: Lecture 02 — Section 1.4.2

Part 2: Essay and Calculation Questions

1. Compare circuit switching and packet switching using the restaurant analogy.

قارن بين تبديل الدوائر وتبديل الحزم باستخدام تشبيه المطاعم.

Model answer:

- Circuit switching: the resources needed along a path (buffers, link transmission rate) are reserved for the duration of the session. Traditional telephone networks are examples.
- Packet switching: resources are not reserved; packets share them on demand and may have to wait.
- Analogy: a restaurant requiring reservations needs a call before leaving home but you can be seated immediately on arrival (circuit switching). A restaurant with no reservations needs no booking but you may wait for a table (packet switching).
- A circuit in a link is implemented with frequency-division multiplexing (FDM) or time-division multiplexing (TDM).

● Ref: Lecture 02 — Section 1.3.2

2. Discuss the arguments for and against packet switching, and state the current trend.

ناقش الحجج المؤيدة والمعارضة لتبديل الحزم، واذكر الاتجاه الحالي.

Model answer:

- Against: critics argue it is unsuitable for real-time services (telephone calls, video conferences) because of variable and unpredictable end-to-end delays, mainly due to queuing delays.
- For: proponents argue that (1) it offers better sharing of transmission capacity than circuit switching, and (2) it is simpler, more efficient, and less costly to implement.
- Trend: both are prevalent today, but the trend is toward packet switching. Even many circuit-switched telephone networks are migrating to it, often using it for the expensive overseas portion of a call.

● Ref: Lecture 02 — Section 1.3.2

3. Explain why connecting each access ISP directly to every other does not scale, and describe Network Structures 1 and 2.

اشرح لماذا لا يتوسع ربط كل access ISP مباشرة بكل الآخرين، وصف Network Structure 1 و2.

Model answer:

- Direct mesh: each access ISP would need a separate link to each of hundreds of thousands of other access ISPs, which is much too costly.
- Network Structure 1: all access ISPs connect to a single (imaginary) global transit ISP, a network of routers and links spanning the globe with a router near each access ISP. The access ISP pays the global ISP and is its customer; the global ISP is the provider.
- Network Structure 2: because the global ISP can be profitable, competitors build their own, giving hundreds of thousands of access ISPs and multiple global transit ISPs. Access ISPs prefer it since they can choose by price and services, but the global transit ISPs must interconnect. It is a two-tier hierarchy.

● Ref: Lecture 02 — Section 1.3.3

4. Describe Network Structure 3 and the customer-provider relationships in it.

صف Network Structure 3 وعلاقات العميل والمزود فيها.

Model answer:

- No ISP has presence in every city; in each region there may be a regional ISP to which the access ISPs connect, and each regional ISP connects to tier-1 ISPs.
- Tier-1 ISPs are similar to the imaginary global transit ISP but do exist (about a dozen, e.g., Level 3 Communications, AT&T, Sprint, NTT) and do not have presence in every city.
- Each access ISP pays its regional ISP, and each regional ISP pays its tier-1 ISP (an access ISP may also connect directly to a tier-1 ISP and pay it). Tier-1 ISPs pay no one.
- In some regions a larger regional ISP (possibly spanning a country) may connect smaller regional ISPs to a tier-1 ISP. This is still a crude approximation of today's Internet.

● Ref: Lecture 02 — Section 1.3.3

5. Define PoP, multi-homing, peering, and IXP, and explain how they form Network Structure 4.

عرّف PoP، multi-homing، peering و IXP و اشرح كيف تشكّل Network Structure 4.

Model answer:

- PoP: a group of one or more routers in the provider's network where customer ISPs can connect into the provider ISP.
- Multi-homing: any ISP except tier-1 may connect to two or more providers, so it can keep sending and receiving packets if one provider fails.
- Peering: a pair of nearby ISPs at the same level connect directly so traffic between them avoids upstream intermediaries, reducing costs. It is typically settlement-free; tier-1 ISPs also peer settlement-free.
- IXP: a meeting point created by a third party, typically a stand-alone building with its own switches, where multiple ISPs can peer.
- Together with access, regional and tier-1 ISPs, these form Network Structure 4.

● Ref: Lecture 02 — Section 1.3.3

6. Describe Network Structure 5 and explain why content providers such as Google build private networks.

صف Network Structure 5 و اشرح لماذا يبني مزودو المحتوى مثل Google شبكات خاصة.

Model answer:

- Network Structure 5 describes today's Internet and adds content-provider networks on top of Network Structure 4.
- Google example: about 19 major data centers across North America, Europe, Asia, South America, and Australia, each with tens or hundreds of thousands of servers, interconnected by Google's private TCP/IP network, which is separate from the public Internet and carries only traffic to/from Google servers.
- The private network tries to bypass upper tiers by peering (settlement-free) with lower-tier ISPs, directly or at IXPs. Since many access ISPs can only be reached through tier-1 networks, it also connects to tier-1 ISPs and pays them for that traffic.
- Benefits: lower payments to upper-tier ISPs and greater control over how services are delivered to end users.

● Ref: Lecture 02 — Section 1.3.3

7. Explain the four types of delay that make up nodal delay in a packet-switched network.

اشرح أنواع التأخير الأربعة التي يتكون منها التأخير عند العقدة في شبكة تبديل الحزم.

Model answer:

- Processing delay: time to examine the packet header and decide where to direct it, plus checking for bit-level errors. Typically microseconds or less in high-speed routers.
- Queuing delay: time waiting in the queue before being transmitted onto the link; depends on the number of earlier-arriving packets. It is zero if the queue is empty and the link is idle; microseconds to milliseconds in practice.
- Transmission delay: L/R , the time to push all of the packet's bits into the link (L = packet length in bits, R = link rate). Microseconds to milliseconds in practice.
- Propagation delay: d/s , the time for a bit to travel across the link (d = distance, s = propagation speed). Milliseconds in wide-area networks.
- Total: $d(\text{nodal}) = d(\text{proc}) + d(\text{queue}) + d(\text{trans}) + d(\text{prop})$.

● Ref: Lecture 02 — Section 1.4.1

8. Distinguish between transmission delay and propagation delay, and explain when each can dominate.

ميّز بين تأخير الإرسال وتأخير الانتشار، واطرح متى يهيمن كل منهما.

Model answer:

- Transmission delay = L/R : the time to push all bits of the packet onto the link; it depends on packet length and link rate (e.g., $R = 10$ Mbps or 100 Mbps Ethernet).
- Propagation delay = d/s : the time for a bit to travel from the beginning of the link to the next router; it depends on distance and the medium's propagation speed (2×10^8 to 3×10^8 m/s, about the speed of light).
- $d(\text{prop})$ can be negligible (a couple of microseconds) for two routers on the same campus, but hundreds of milliseconds over a geostationary satellite link, where it can be the dominant term in $d(\text{nodal})$.
- $d(\text{trans})$ can range from negligible to significant.

● Ref: Lecture 02 — Section 1.4.1

9. Explain traffic intensity and how it affects the queuing delay, including the effect of bursty arrivals.

اشرح شدة الحركة (traffic intensity) وتأثيرها على تأخير الطابور، مع تأثير الوصول على شكل دفعات.

Model answer:

- Let λ be the average packet arrival rate (packets/sec), L the packet size in bits and R the transmission rate. Traffic intensity = $\lambda L / R$.
- If $\lambda L / R > 1$, bits arrive faster than they can be sent, the queue grows without bound and the delay approaches infinity. Golden rule: design so the traffic intensity is no greater than 1.
- If $\lambda L / R \leq 1$, the nature of the traffic matters: if one packet arrives every L/R seconds, each finds an empty queue and there is no queuing delay. If N packets arrive together every $(L/R)N$ seconds, the n th transmitted packet waits $(n-1)L/R$.
- If the traffic intensity is close to zero, arrivals are rare and the average queuing delay is close to zero. As it approaches 1, the average queuing delay increases rapidly.

● Ref: Lecture 02 — Section 1.4.2

10. Explain packet loss, and describe the end-to-end delay equation and the role of Traceroute.

اشرح فقد الحزم، وصف معادلة التأخير من طرف إلى طرف ودور برنامج Traceroute.

Model answer:

- Packet loss: the queue before a link has finite capacity, so an arriving packet may find it full and the router drops it. The fraction of lost packets increases with traffic intensity, so node performance is measured by delay and by probability of loss. A lost packet may be retransmitted end-to-end.
- End-to-end delay: with $N-1$ routers, an uncongested network (negligible queuing), the same $d(\text{proc})$, R and $d(\text{prop})$ at each node/link: $d(\text{end-to-end}) = N (d(\text{proc}) + d(\text{trans}) + d(\text{prop}))$, with $d(\text{trans}) = L/R$.
- This generalizes Equation 1.1, $d = N L/R$, which ignored processing and propagation delays.
- Traceroute: a simple program that can run in any Internet host; the user gives a destination hostname and the program in the source host sends multiple special packets toward that destination, giving a hands-on feel for end-to-end delay.

● Ref: Lecture 02 — Section 1.4.2

11. Calculate the transmission delay of a packet of 1,500 bytes on (a) a 10 Mbps Ethernet link and (b) a 100 Mbps Ethernet link.

احسب تأخير الإرسال لحزمة حجمها 1,500 بايت على (أ) رابط Ethernet سرعته 10 Mbps و(ب) رابط سرعته 100 Mbps.

Model answer:

- Given: $L = 1,500 \times 8 = 12,000$ bits.
- Formula: $d(\text{trans}) = L/R$.
- (a) $R = 10$ Mbps: $12,000 / 10,000,000 = 0.0012$ s = 1.2 ms.
- (b) $R = 100$ Mbps: $12,000 / 100,000,000 = 0.00012$ s = 0.12 ms.
- The 100 Mbps link is 10 times faster, so the delay is 10 times smaller.

● Ref: Lecture 02 — Section 1.4.1

12. Two routers are connected by a link of length 2,500 km, and the propagation speed of the link is 2.5×10^8 m/s. Calculate the propagation delay.

راوتران بينهما رابط طوله 2,500 كم وسرعة الانتشار فيه 2.5×10^8 م/ث. احسب تأخير الانتشار.

Model answer:

- Given: $d = 2,500$ km = 2.5×10^6 m; $s = 2.5 \times 10^8$ m/s.
- Formula: $d(\text{prop}) = d/s$.
- $d(\text{prop}) = (2.5 \times 10^6) / (2.5 \times 10^8) = 0.01$ s.
- $d(\text{prop}) = 10$ ms.

● Ref: Lecture 02 — Section 1.4.1

13. A router's outgoing link has rate $R = 1$ Mbps. Packets have length $L = 1,000$ bits. Compute the traffic intensity and comment on the queue when (a) the average arrival rate is $a = 800$ packets/sec and (b) $a = 1,200$ packets/sec.

رابط خارج من راوتر معدله $R = 1$ Mbps وطول الحزم $L = 1,000$ bit. احسب شدة الحركة وعلّق على الطابور عندما (أ) $a = 800$ حزمة/ث و(ب) $a = 1,200$ حزمة/ث.

Model answer:

- Formula: traffic intensity = La/R .
- (a) $La/R = (1,000 \times 800) / 1,000,000 = 0.8$. This is less than 1, so the queue does not grow without bound, although the average queuing delay is noticeable because the intensity is close to 1.
- (b) $La/R = (1,000 \times 1,200) / 1,000,000 = 1.2$. This is greater than 1: bits arrive faster than they can be transmitted, so the queue grows without bound, the queuing delay approaches infinity and packets will eventually be dropped when the finite buffer fills.
- Golden rule: design the system so the traffic intensity is no greater than 1.

● Ref: Lecture 02 — Section 1.4.2

14. Ten packets, each of $L = 12,000$ bits, arrive at the same time at an empty queue in front of a link with $R = 10$ Mbps. Find the queuing delay of the 5th and the 10th transmitted packets.

وصلت 10 حزم، كل منها $L = 12,000$ bit، في نفس اللحظة إلى طابور فارغ أمام رابط معدله $R = 10$ Mbps. أوجد تأخير الطابور للحزمة الخامسة والعاشر.

Model answer:

- Given: $L = 12,000$ bits; $R = 10,000,000$ bits/s.
- $L/R = 12,000 / 10,000,000 = 0.0012$ s = 1.2 ms.
- Formula: the n th transmitted packet has queuing delay $(n-1) L/R$.
- 5th packet: $(5-1) \times 1.2$ ms = 4.8 ms.
- 10th packet: $(10-1) \times 1.2$ ms = 10.8 ms (it waits for the other nine packets).

● Ref: Lecture 02 — Section 1.4.2

15. A path has $N = 3$ links (2 routers) and the network is uncongested. Each node has $d(\text{proc}) = 0.1$ ms, each link has rate $R = 2$ Mbps and length 1,000 km with propagation speed 2×10^8 m/s. A packet of $L = 8,000$ bits is sent. Compute the end-to-end delay.

مسار فيه $N = 3$ روابط (راوترين) والشبكة غير مزدحمة. تأخير المعالجة عند كل عقدة 0.1 ms، ومعدل كل رابط $R = 2$ Mbps وطوله 1,000 كم وسرعة الانتشار 2×10^8 م/ث. أرسلت حزمة $L = 8,000$ bit. احسب التأخير من طرف إلى طرف.

Model answer:

- Given: $N = 3$; $d(\text{proc}) = 0.1$ ms; $L = 8,000$ bits; $R = 2,000,000$ bits/s; $d = 1,000$ km = 10^6 m; $s = 2 \times 10^8$ m/s; queuing is negligible.
- Formula (Equation 1.2): $d(\text{end-to-end}) = N (d(\text{proc}) + d(\text{trans}) + d(\text{prop}))$.
- $d(\text{trans}) = L/R = 8,000 / 2,000,000 = 0.004$ s = 4 ms.
- $d(\text{prop}) = d/s = 10^6 / (2 \times 10^8) = 0.005$ s = 5 ms.
- $d(\text{end-to-end}) = 3 \times (0.1 + 4 + 5) = 3 \times 9.1 = 27.3$ ms.

● Ref: Lecture 02 — Section 1.4.2